

Matrix definition and printing

$$A = \begin{pmatrix} 1 & -1 & -2 \\ 1 & 0 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & -2 \end{pmatrix}, D = \begin{pmatrix} 8 & 4 & 0 \\ 4 & -1 & 5 \\ 1 & -1 & 1 \end{pmatrix}, I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Matrix addition - Linear combination $C = A + B = \begin{pmatrix} 1 & -1 & -2 \\ 1 & 0 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & -2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -2 \\ 2 & 3 & 1 \end{pmatrix}$

$$J = 2A - 3B = 2 \begin{pmatrix} 1 & -1 & -2 \\ 1 & 0 & 3 \end{pmatrix} - 3 \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & -2 \end{pmatrix} = \begin{pmatrix} -1 & -8 & -4 \\ -1 & -9 & 12 \end{pmatrix}$$

$$K = A - B = \begin{pmatrix} 1 & -1 & -2 \\ 1 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & -2 \end{pmatrix} = \begin{pmatrix} 0 & -3 & -2 \\ 0 & -3 & 5 \end{pmatrix}$$

Matrix multiplication $H = A \cdot D = \begin{pmatrix} 1 & -1 & -2 \\ 1 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 8 & 4 & 0 \\ 4 & -1 & 5 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 7 & -7 \\ 11 & 1 & 3 \end{pmatrix}$

Transpose matrix $H_T = \begin{pmatrix} 2 & 7 & -7 \\ 11 & 1 & 3 \end{pmatrix}^T = \begin{pmatrix} 2 & 11 \\ 7 & 1 \\ -7 & 3 \end{pmatrix}$

Trace - Determinant Matrix D is Square. The trace is $\text{tr}D = 8$. The determinant is

$$\det D = \begin{vmatrix} 8 & 4 & 0 \\ 4 & -1 & 5 \\ 1 & -1 & 1 \end{vmatrix} = 36$$

Matrix power $F = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 4 & -2 \\ 3 & 1 & 0 \end{pmatrix} \Rightarrow G = F^2 = \begin{pmatrix} -2 & 9 & -5 \\ -6 & 14 & -8 \\ 3 & 10 & -5 \end{pmatrix}$